SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofArtificialIntelligenceandDataScience

ASSESSMENTTOOL1 –ANALYTICALPROBLEMSOLVING

CourseCode:	DSA03	**CourseTitle:**	NaturalLanguageProcessing
COAssessed:	CO2 – Manipulation	**Assessment:**	AssessmentTool1–ANALYTIC AL PROBLEM SOLVING
Weightage:	50% ofCIA		
CO5Statement:	Designinflectional,derivational,finite-stateparsing,stemmingalgorithmsandforanal yse morphological methods. (BL-3)		
Student Name:	Y. Harsha Vardhan Reddy	**Reg. No.:**	192425227
Section/Batch:	B.TechAI-ML	**Date:**	29-07-2026

CodeofConduct

I Y.Harsha Vardhan Reddy (Name /Reg No) certify that thissubmission is myoriginalworkand thatIhaveadheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrity rules will result in disciplinary action.

Signature of theStudent:

Instructions

- Readallquestionscarefullybeforeattemptingtheanswers.
- Answerallfivequestions.Eachquestioncarriesequalmarks.
- Showallintermediatesteps,calculations,probabilitycomputations,taggingdecisions,andjustifications wherever applicable.
- Duration:30minutes.

EnglishWordClasses,PartofSpee ch Tagging	Morphologicaldecompositionofconnected,connecting , and connection; identification of roots, suffixes, inflectionalvs.derivationalforms,andnormalizationfor document indexing.	Q1
EnglishWordClasses,TagsetsforEngli sh, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
CountingWords,PartofSpeechTaggin g, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-StateMorphologicalAnalysis,R ule Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modelingregularandirregular verb inflections androot normalization.	Q4
EnglishWordClasses,Rule-BasedPar tof Speech Tagging, Counting Words	Porter Stemmer-based processing ofrelational,relation, andrelate;applicationofderivationalsuffixremovaland stem extraction for retrieval.	Q5

Q1.MorphologicalAnalysisPipeline

InputWords:

- Connected
- Connecting
- Connection

Step1:DecomposeintoRootandSuffix

Word	Root	Suffix
Connected	Connect	-ed
Connecting	Connect	-ing
Connection	Connect	-ion

Step2:Classification

- **Connected**→Inflectional(Pasttense)
- **Connecting**→Inflectional(Presentparticiple)
- **Connection**→Derivational(Formsanoun)

Step3:Normalization

Allwordsareconvertedtothecommonbaseform:

Connect

ParsedStructure

- Connected=Connect+ ed
- Connecting=Connect+ ing
- Connection=Connect+ ion

FinalOutput

NormalizedRoot=Connect

Q2.MorphologicalParsingModule

InputWords

- Unhappy
- Happiness
- Happily

Step1:MorphologicalBreakdown

Word		BaseWord	Suffix

Unhappy	un-	happy	—
Happiness		happy	-ness
Happily	—	happy	-ly

Step2:Classification

- Unhappy→Derivational
- Happiness→Derivational
- Happily→Derivational

SemanticEffect

- **un-**changesmeaningtoopposite.
- **-ness**changes adjectiveintonoun.
- **-ly**changes adjectiveintoadverb.

FinalRoot

Happy

Q3.Stemming-BasedPreprocessing

InputWords

- Played
- Player
- Playing

Step1:RemoveAffixes

Word		Affix
Played	Play	-ed
Player	Play	-er
Playing	Play	-ing

Step2:Classification

- Played→Inflectional
- Player→Derivational
- Playing → Inflectional

FinalStem

Play

Result
Allwordsareindexed as:
Play

Q4.Finite-StateMorphologicalParsing

InputWords

- Writes
- Writing
- Written

MorphologicalBreakdown

Word	Root	Suffix	Type
Write s	Write	-s	RegularInflection
Writing	Write	-ing	RegularInflection
Written	Write	-en	IrregularInflection

StateTransitions
StartState→ Root (**Write**) · Write→ +s→ Writes

- Write→ +ing→ Writing
- Write→+en→Written

NormalizedRoot
Write
Result
Allformsmap to:
Write

Q5.PorterStemmer-BasedProcessing

InputWords

- Relational
- Relation
- Relate

Step-by-StepStemming

Word	RemoveSuffix	IntermediateForm	FinalStem
Relational	-ational	Relate	Relat
Relation	-ion	Relat	Relat

Relate	-e	Relat	Relat

Explanation

- Removederivationalsuffixes.
- ApplyPorterStemmerrules.
- Convertallwordstoone commonstem.

FinalStem

Relat

Result

Allwordsareindexed as:

Relat

Conclusion

Morphologicalanalysisandstemminghelpconvertdifferentformsofawordintoacommonrootorstem. This improves document indexing, searching, and information retrieval by treating related words as the same term.

RubricsforCalculation

NumericalProblems

Criteria	Weight ag e	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understa nding	20%	Clearly interprets problem,identifi es allparametersa nd requirements accurately	Understands problemwithmi nor gaps	Partial understanding; misses key elements	Misinterpr ets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct methodwithmin or errors	Inappropriateor partially correct method	Unable to selectmethod
Solution Process	25%	Logical,systema tic steps with correct approach throughout	Mostly correct stepswithminor mistakes	Disorganizedste ps; multiple errors	Noclear process
Accura cy of Calcula	20%	All calculations accurate;corr ect final	Minorcalculati on errors	Frequenterrors; incorrect final answer	No valid calculations

tio n		answer			
Interpr eta tion of Results	15%	Clearlyinterpr ets resultswithunit s and engineering relevance	Basic interpretationw ith minor omissions	Limitedoruncle ar interpretation	No interpretation

Q.No./ Criter ia Level	Problem Understanding	Method Selection	Soluti on Proce ss	Accuracy of Calculati on	Interpretati on of Results	Sub. Tota l	Total %
1							
2							
3							
4							
5							

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date: